

MPG No. 7.3.22 $m'm2'$ (m'm2' setting) [Type III, orthorhombic] [M tensor]

* Rank 0 tensor. * Rank 1 tensor.

$$\begin{bmatrix} 0 & M_y & 0 \end{bmatrix}$$

$$M_y = M_{py}^{(1)}$$

* Rank 2 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & M_{yz} \\ 0 & M_{yz} & 0 \end{bmatrix}$$

$$M_{yz} = M_{dyz}^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & M_{yz} \\ 0 & -M_{yz} & 0 \end{bmatrix}$$

$$M_{yz} = T_{px}^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & M_{xxy} & 0 \\ 0 & M_{yyy} & 0 \\ 0 & M_{zzy} & 0 \\ 0 & 0 & M_{zzz} \\ 0 & 0 & 0 \\ M_{xxy} & 0 & 0 \end{bmatrix}$$

$$M_{xxy} = -M_{fay}^{(1)} - M_{fby}^{(1)} + M_{py}^{(1)}$$

$$M_{yyy} = 2M_{fay}^{(1)} + 3M_{py}^{(1)}$$

$$M_{zzz} = -M_{fay}^{(1)} + M_{fby}^{(1)} + M_{py}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & M_{xxy} & 0 \\ 0 & 0 & 0 \\ 0 & M_{zzy} & 0 \\ 0 & 0 & -\frac{M_{zzz}}{2} \\ 0 & 0 & 0 \\ -\frac{M_{xxy}}{2} & 0 & 0 \end{bmatrix}$$

$$M_{xxy} = 2M_{py}^{(2)} - 2T_{dxz}^{(1)}$$

$$M_{zzy} = 2M_{py}^{(2)} + 2T_{dxz}^{(1)}$$

$$\frac{M_{zzz}}{2} = M_{py}^{(2)} + T_{dxz}^{(1)}$$

$$\frac{M_{xxy}}{2} = M_{py}^{(2)} - T_{dxz}^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} 0 & 0 & M_{yzz} \\ 0 & 0 & 0 \\ M_{xyx} & 0 & 0 \end{bmatrix}$$

$$M_{yzz} = -M_{py}^{(3)} + T_{dxz}^{(2)}$$

$$M_{xyx} = M_{py}^{(3)} + T_{dxz}^{(2)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xyz} & 0 & 0 \\ 0 & 0 & 0 & M_{yyy} & 0 & 0 \\ 0 & 0 & 0 & M_{zzz} & 0 & 0 \\ M_{xyz} & M_{yyy} & M_{zzz} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{zxy} \\ 0 & 0 & 0 & 0 & M_{zxy} & 0 \end{bmatrix}$$

$$M_{xyz} = M_{dyz}^{(1)} + 2M_{gbx}^{(1)}$$

$$M_{yyy} = M_{dyz}^{(1)} + 2M_{dyz}^{(2)} + M_{gax}^{(1)} - M_{gbx}^{(1)}$$

$$M_{zzz} = M_{dyz}^{(1)} + 2M_{dyz}^{(2)} - M_{gax}^{(1)} - M_{gbx}^{(1)}$$

$$M_{zxy} = M_{dyz}^{(2)} + 2M_{gbx}^{(1)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xyz} & 0 & 0 \\ 0 & 0 & 0 & M_{yyy} & 0 & 0 \\ 0 & 0 & 0 & M_{zzz} & 0 & 0 \\ -M_{xyz} & -M_{yyy} & -M_{zzz} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{zxy} \\ 0 & 0 & 0 & 0 & -M_{zxy} & 0 \end{bmatrix}$$

$$M_{xyz} = M_{dyz}^{(3)} - 4T_{fbx}^{(1)}$$

$$M_{yyy} = M_{dyz}^{(3)} - 2T_{fax}^{(1)} + 2T_{fbx}^{(1)} + 2T_{px}^{(1)}$$

$$M_{zzz} = M_{dyz}^{(3)} + 2T_{fax}^{(1)} + 2T_{fbx}^{(1)} - 2T_{px}^{(1)}$$

$$M_{zxy} = -4T_{fax}^{(1)} - T_{px}^{(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & M_{zxy} \\ 0 & M_{zxy} & 0 \end{bmatrix}$$

$$M_{zxy} = 2M_{dyz}^{(4)}$$

* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & M_{zxy} \\ 0 & -M_{zxy} & 0 \end{bmatrix}$$

$$M_{zxy} = -T_{px}^{(2)}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} M_{xyz} & 0 & 0 \\ M_{yyy} & 0 & 0 \\ M_{zzz} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & M_{zxy} \\ 0 & M_{xyz} & 0 \end{bmatrix}$$

$$\begin{aligned}
M_{xxyz} &= 2T_{fax}^{(2)} + T_{px}^{(3)} + 2T_{px}^{(4)} \\
M_{yyyy} &= -2M_{dyz}^{(5)} - T_{fax}^{(2)} + T_{fbx}^{(2)} + T_{px}^{(3)} \\
M_{zzyz} &= 2M_{dyz}^{(5)} - T_{fax}^{(2)} - T_{fbx}^{(2)} + T_{px}^{(3)} \\
M_{zxyy} &= -M_{dyz}^{(5)} - T_{fax}^{(2)} - T_{fbx}^{(2)} + T_{px}^{(4)} \\
M_{xyzx} &= M_{dyz}^{(5)} - T_{fax}^{(2)} + T_{fbx}^{(2)} + T_{px}^{(4)}
\end{aligned}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} M_{yzxx} & M_{yzyy} & M_{yzzz} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{zxxxy} \\ 0 & 0 & 0 & 0 & M_{xyzx} & 0 \end{bmatrix}$$

$$\begin{aligned}
M_{yzxx} &= 2T_{fax}^{(3)} + T_{px}^{(5)} + 2T_{px}^{(6)} \\
M_{yzyy} &= -2M_{dyz}^{(6)} - T_{fax}^{(3)} + T_{fbx}^{(3)} + T_{px}^{(5)} \\
M_{yzzz} &= 2M_{dyz}^{(6)} - T_{fax}^{(3)} - T_{fbx}^{(3)} + T_{px}^{(5)} \\
M_{zxxxy} &= M_{dyz}^{(6)} - T_{fax}^{(3)} + T_{fbx}^{(3)} + T_{px}^{(6)} \\
M_{xyzx} &= -M_{dyz}^{(6)} - T_{fax}^{(3)} - T_{fbx}^{(3)} + T_{px}^{(6)}
\end{aligned}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xxyz} & 0 & 0 & M_{xxzy} & 0 & 0 \\ 0 & 0 & 0 & M_{yyyz} & 0 & 0 & M_{yyzy} & 0 & 0 \\ 0 & 0 & 0 & M_{zzyz} & 0 & 0 & M_{zzzy} & 0 & 0 \\ M_{yzxx} & M_{yzyy} & M_{yzzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{zxxxy} & 0 & 0 & M_{zxyxx} \\ 0 & 0 & 0 & 0 & M_{xyzx} & 0 & 0 & M_{xyxz} & 0 \end{bmatrix}$$

$$\begin{aligned}
M_{xxyz} &= M_{dyz}^{(1)} + M_{dyz}^{(3)} + 2M_{gbx}^{(1)} + 2T_{fax}^{(2)} - 4T_{fbx}^{(1)} + T_{px}^{(3)} + 2T_{px}^{(4)} \\
M_{xxzy} &= M_{dyz}^{(1)} + M_{dyz}^{(3)} + 2M_{gbx}^{(1)} - 2T_{fax}^{(2)} - 4T_{fbx}^{(1)} - T_{px}^{(3)} - 2T_{px}^{(4)} \\
M_{yyyy} &= M_{dyz}^{(1)} + 2M_{dyz}^{(2)} + M_{dyz}^{(3)} - 2M_{dyz}^{(5)} + M_{gax}^{(1)} - M_{gbx}^{(1)} - 2T_{fax}^{(1)} - T_{fax}^{(2)} + 2T_{fbx}^{(1)} + T_{fbx}^{(2)} + 2T_{px}^{(1)} + T_{px}^{(3)} \\
M_{yyzy} &= M_{dyz}^{(1)} + 2M_{dyz}^{(2)} + M_{dyz}^{(3)} + 2M_{dyz}^{(5)} + M_{gax}^{(1)} - M_{gbx}^{(1)} - 2T_{fax}^{(1)} + T_{fax}^{(2)} + 2T_{fbx}^{(1)} - T_{fbx}^{(2)} + 2T_{px}^{(1)} - T_{px}^{(3)} \\
M_{zzyz} &= M_{dyz}^{(1)} + 2M_{dyz}^{(2)} + M_{dyz}^{(3)} + 2M_{dyz}^{(5)} - M_{gax}^{(1)} - M_{gbx}^{(1)} + 2T_{fax}^{(1)} - T_{fax}^{(2)} + 2T_{fbx}^{(1)} - T_{fbx}^{(2)} - 2T_{px}^{(1)} + T_{px}^{(3)} \\
M_{zzzy} &= M_{dyz}^{(1)} + 2M_{dyz}^{(2)} + M_{dyz}^{(3)} - 2M_{dyz}^{(5)} - M_{gax}^{(1)} - M_{gbx}^{(1)} + 2T_{fax}^{(1)} + T_{fax}^{(2)} + 2T_{fbx}^{(1)} + T_{fbx}^{(2)} - 2T_{px}^{(1)} - T_{px}^{(3)} \\
M_{yzxx} &= M_{dyz}^{(1)} - M_{dyz}^{(3)} + 2M_{gbx}^{(1)} + 4T_{fbx}^{(1)} \\
M_{yzyy} &= M_{dyz}^{(1)} + 2M_{dyz}^{(2)} - M_{dyz}^{(3)} + M_{gax}^{(1)} - M_{gbx}^{(1)} + 2T_{fax}^{(1)} - 2T_{fbx}^{(1)} - 2T_{px}^{(1)} \\
M_{yzzz} &= M_{dyz}^{(1)} + 2M_{dyz}^{(2)} - M_{dyz}^{(3)} - M_{gax}^{(1)} - M_{gbx}^{(1)} - 2T_{fax}^{(1)} - 2T_{fbx}^{(1)} + 2T_{px}^{(1)} \\
M_{zxxxy} &= M_{dyz}^{(2)} - M_{dyz}^{(5)} + 2M_{gbx}^{(1)} - 4T_{fax}^{(1)} - T_{fax}^{(2)} - T_{fbx}^{(2)} - T_{px}^{(1)} + T_{px}^{(4)} \\
M_{zxyx} &= M_{dyz}^{(2)} + M_{dyz}^{(5)} + 2M_{gbx}^{(1)} - 4T_{fax}^{(1)} + T_{fax}^{(2)} + T_{fbx}^{(2)} - T_{px}^{(1)} - T_{px}^{(4)} \\
M_{xyzx} &= M_{dyz}^{(2)} + M_{dyz}^{(5)} + 2M_{gbx}^{(1)} + 4T_{fax}^{(1)} - T_{fax}^{(2)} + T_{fbx}^{(2)} + T_{px}^{(1)} + T_{px}^{(4)} \\
M_{xyxz} &= M_{dyz}^{(2)} - M_{dyz}^{(5)} + 2M_{gbx}^{(1)} + 4T_{fax}^{(1)} + T_{fax}^{(2)} - T_{fbx}^{(2)} + T_{px}^{(1)} - T_{px}^{(4)}
\end{aligned}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} M_{yzxx} & M_{yzyy} & M_{yzzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{zxxxy} & 0 & 0 & M_{zxyxx} \\ 0 & 0 & 0 & 0 & M_{xyzx} & 0 & 0 & M_{xyxz} & 0 \end{bmatrix}$$

$$\begin{aligned}
M_{yzxx} &= 2T_{fax}^{(3)} + T_{px}^{(5)} + 2T_{px}^{(6)} \\
M_{yzyy} &= -2M_{dyz}^{(6)} - T_{fax}^{(3)} + T_{fbx}^{(3)} + T_{px}^{(5)} \\
M_{yzzz} &= 2M_{dyz}^{(6)} - T_{fax}^{(3)} - T_{fbx}^{(3)} + T_{px}^{(5)} \\
M_{zxxy} &= 2M_{dyz}^{(4)} + M_{dyz}^{(6)} - T_{fax}^{(3)} + T_{fbx}^{(3)} - T_{px}^{(2)} + T_{px}^{(6)} \\
M_{zxyx} &= -2M_{dyz}^{(4)} + M_{dyz}^{(6)} - T_{fax}^{(3)} + T_{fbx}^{(3)} + T_{px}^{(2)} + T_{px}^{(6)} \\
M_{xyzx} &= 2M_{dyz}^{(4)} - M_{dyz}^{(6)} - T_{fax}^{(3)} - T_{fbx}^{(3)} + T_{px}^{(2)} + T_{px}^{(6)} \\
M_{xyxz} &= -2M_{dyz}^{(4)} - M_{dyz}^{(6)} - T_{fax}^{(3)} - T_{fbx}^{(3)} - T_{px}^{(2)} + T_{px}^{(6)}
\end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix}
0 & 0 & 0 \\
0 & 0 & M_{yyyz} \\
0 & M_{zzzy} & 0 \\
0 & M_{yyzy} & 0 \\
0 & 0 & 0 \\
0 & 0 & M_{xxyz} \\
0 & 0 & M_{yzzz} \\
0 & M_{zxxxy} & 0 \\
0 & 0 & 0 \\
-\frac{M_{yyyyz}}{2} + \frac{M_{yyzyy}}{2} + M_{zxxxy} & 0 & 0
\end{bmatrix}$$

$$\begin{aligned}
M_{yyyz} &= M_{dyz}^{(1)} + 2M_{dyz}^{(2)} + M_{dyz}^{(3)} - 2M_{dyz}^{(5)} + M_{gax}^{(1)} - M_{gbx}^{(1)} \\
M_{zzzy} &= M_{dyz}^{(1)} + 2M_{dyz}^{(2)} + M_{dyz}^{(3)} - 2M_{dyz}^{(5)} - M_{gax}^{(1)} - M_{gbx}^{(1)} \\
M_{yyzy} &= M_{dyz}^{(1)} + 2M_{dyz}^{(2)} + M_{dyz}^{(3)} + 2M_{dyz}^{(5)} + M_{gax}^{(1)} - M_{gbx}^{(1)} \\
M_{xxyz} &= M_{dyz}^{(1)} + M_{dyz}^{(3)} + 2M_{gbx}^{(1)} \\
M_{yzzz} &= M_{dyz}^{(1)} + 2M_{dyz}^{(2)} - M_{dyz}^{(3)} - M_{gax}^{(1)} - M_{gbx}^{(1)} \\
M_{zxxxy} &= M_{dyz}^{(2)} - M_{dyz}^{(5)} + 2M_{gbx}^{(1)}
\end{aligned}$$